Credit Name: CSE 2140 2nd Language Programming

Assignment Name: change.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

I read through the excercise to undestand that the goal was to take a coin amount in cents and break it down into the minimum possible number of quarters, dimes, nickels, and pennies. I looked at the sample output aswell

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Tracking how the remaining money changes after removing each coin denomination was a bit tricky at first but i resolved this by using compound assignment operators like %= to carry over the remainder cleanly to the next coin type

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes i broke it down, i started with the largest coin being quarters, finding how many fit, keeping the remainder and repeating the process down to the pennies

How did you decide on the tools, data structures, or algorithms to use?

used int variable for all coin calculations because coin counts and cents are whole numbers. i used division to find the count and the modulus operator to find what money was left over

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

Yes, step by step. I first set up the Scanner to accept user input, then wrote the math for quarters and tested it before moving down to dimes, nickels and pennies

How did you test your solution along the way to make sure it was working?

I ran the program using the sample input 212 from the textbook to verify that my outputs matched the expected count for quarters, dimes, nickles, and pennies

Overcoming Challenges

What part of the problem was the most difficult for you?

making sure the remainder flowed correctly from one coin type to the next without losing track of the total value

How did you handle moments when you felt stuck or unsure of what to do next?

Asked my desk mate once again...

Learning

Was there anything you learned that you think will help you with future challenges?

I learned how efficient the modulus operator is when dealing with currency conversions or scenarios where you need to track remainders after a division